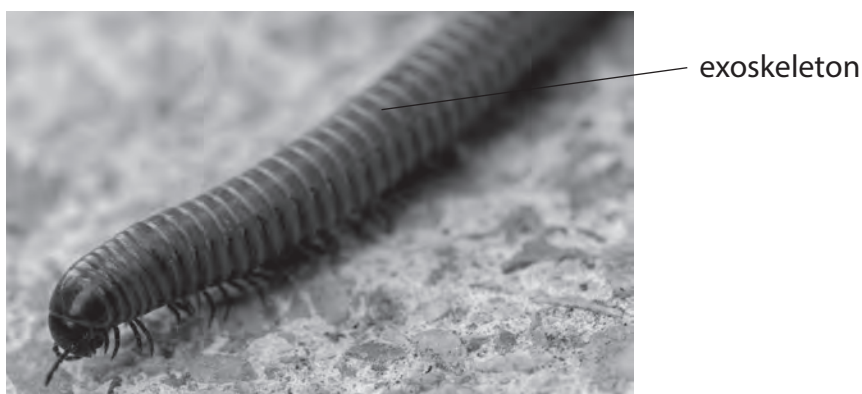


6 Different species of millipede have different adaptations.

One shared adaptation is a hard outer layer called an exoskeleton.

The photograph shows a millipede.

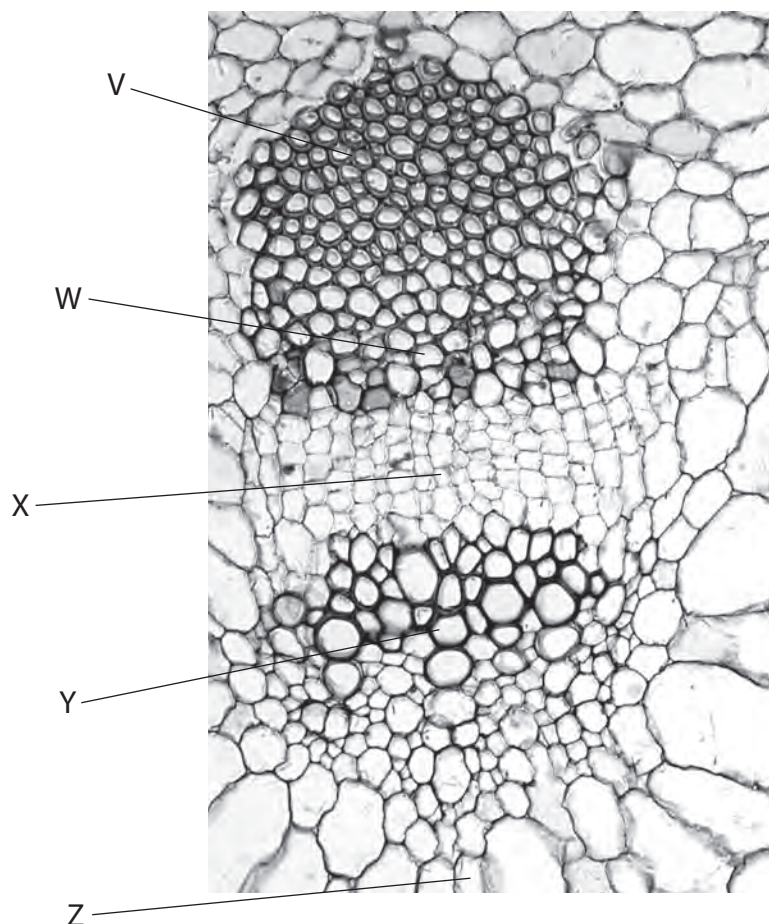


(Source: © Marina Kovacevic / Alamy Stock Photo)

- (a) *Illacme plenipes* is a species of millipede that is endemic to California. It was thought to be extinct but was rediscovered in 2005.

The mouth of this millipede is adapted to make holes in xylem and phloem tissues of living plants.

The photograph shows some plant tissues, as seen using a microscope.



(Source: © Choksawatdikorn / Shutterstock)

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(i) Which of the labelled structures is a xylem vessel?

(1)

☐ **A** W

☐ **B** X

☐ **C** Y

☐ **D** Z

(ii) Which of the labelled structures contains sieve plates?

(1)

☐ **A** V

☐ **B** W

☐ **C** X

☐ **D** Y

(iii) Explain the advantages for this millipede of being able to make holes in xylem and phloem.

(3)

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(b) The *Illacme tobini* species of millipede is also endemic to California.

(i) It has an exoskeleton and 200 poison glands.

Suggest why this millipede has these adaptations.

(2)

Hard exoskeleton

Poison glands

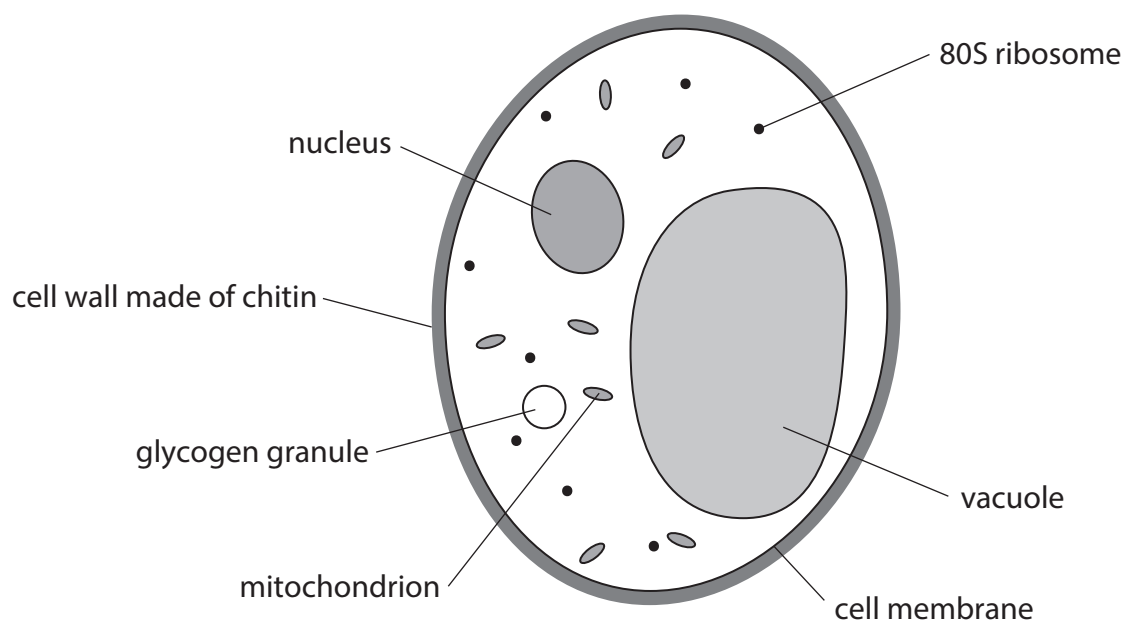
(ii) Explain how scientists could determine how closely related *I. plenipes* is to *I. tobini*.

(2)



(c) Some millipedes have bacteria in their gut that can break down fungal cells.

The diagram shows a fungal cell.



Compare and contrast the structure of this fungal cell and a bacterial cell.

(3)

(Total for Question 6 = 12 marks)